

Mohammad Mahdi Salmani-Zarchi

M.Sc. Student in Artificial Intelligence, University of Tehran

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Research Interests

Natural Language Processing, Reinforcement Learning, LLM-Driven Social Simulations, Instruction Following, Theory of Mind in LLMs, Personalized Agents, Interpretability in Foundation Models.

Skills

AI & ML: Machine Learning, Deep Learning, Statistical Inference, Reinforcement Learning
NLP & LLMs: PEFT (LoRA, QLoRA), RLHF, RAG, Natural Language Inference, Agentic LLMs
Frameworks & Systems: PyTorch, TensorFlow, vLLM, Distributed Training, MLflow, Optuna
Programming & Tools: Python, C++, JavaScript; Git, $\text{\LaTeX}$, Docker

Education



M.Sc. in Computer Engineering - Artificial Intelligence

GPA: 17.95 / 20

University of Tehran, Tehran, Iran

Sep 2023 – Present

- **Thesis:** The Impact of Human Preference Alignment on Large Language Models Role-Playing



B.Sc. in Electrical Engineering

GPA: 15.28 / 20

Isfahan University of Technology, Isfahan, Iran

Sep 2018 – Dec 2022

- **Final Project:** Android app for diagnosing musculoskeletal abnormalities

Publications

* indicates first author

MDP-GRPO: Stabilized Group Relative Policy Optimization for Multi-Constraint Instruction Following* **ACL 2026 Main Conference. DOI**

- Identified and formalized three failure modes of GRPO under discrete, low-dispersion reward settings: low-variance amplification, mean-centering blindness, and zero-variance collapse.
- Proposed MDP-GRPO, a stabilized RL framework for multi-constraint instruction following with verifiable rewards, improving training stability and achieving higher constraint satisfaction.

From Neurotypical to Neurodiverse: Evaluating LLMs' Ability to Simulate Autistic Theory of Mind* Under review at **Computers in Human Behavior** (Q1 Journal). **Preprint**

- Proposed a DSM-5-grounded evaluation framework for assessing theory-of-mind reasoning in LLMs under autism-spectrum-inspired cognitive constraints.
- Analyzed how prompting strategies and alignment mechanisms affect the separability and consistency of simulated cognitive profiles across multiple ToM task types and statistical properties.

Work Experience

Natural Language Processing Engineer

MCI, Tehran, Iran

MCI Lab, R&D Department

March 2025 – March 2026

- Developed and deployed resource-efficient LLM pipelines using supervised fine-tuning and reinforcement learning to improve instruction-following performance, including vLLM-based inference and serving. Used MLflow for experiment tracking.

Research Intern, ODCL Project

ESSEC Business School, Cergy, France

Supervisors: *Prof. Emiliano Traversi*, *Dr. Pegah Alizadeh*, *Dr. Massinissa Hamidi*

May 2026 – Present

- Developed a dynamic curriculum learning framework that adapts task ordering during training via learnable edge-preference parameters, formulated as a graph-based optimization problem.

Research Assistant, NLP Lab

Supervisors: [Prof. Hesham Faili](#), [Dr. M.J. Dousti](#)

University of Tehran, Tehran, Iran

Dec. 2024 – Present

- Conducted research on human preference alignment and role-playing in LLMs, contributing to experimental validation and peer-reviewed manuscripts.

Research Assistant, Cognitive Science Lab

Supervisor: [Dr. Abdol-Hossein Vahabie](#)

University of Tehran, Tehran, Iran

Spring & Summer 2025

- Investigated theory-of-mind and social reasoning capabilities of LLMs using cognitively inspired experimental designs grounded in psychological and DSM-5 frameworks.

Teaching Assistant

Natural Language Processing

Teacher: [Prof. Hesham Faili](#)

University of Tehran, Iran

[GitHub page](#) - Fall 2025

- Designed a course project and hands-on workshop on fine-tuning LLMs using LoRA/PEFT, covering data preparation, tokenizer analysis, training, and standardized evaluation.

Large Language Models

Teachers: [Dr. M.J. Dousti](#), [Dr. Yadollah Yaghoobzadeh](#)

University of Tehran, Iran

Spring 2025

- Developed assignments on in-context learning and human preference alignment, introducing students to prompt engineering and reinforcement learning from human feedback.

Deep Learning

Teacher: [Dr. Ahmad Kalhor](#)

University of Tehran, Iran

Fall 2024, Spring 2025

- Designed assignments on image captioning, focusing on CNN-RNN architectures and attention mechanisms.

Scholarships & Academic Service

- **MCI Scholarship:** Competitive scholarship supporting M.Sc. research in AI.
- **Session Chair:** ACL 2026 Conference, Virtual Session.

Projects

- **Text2SQL Chatbot:** LLM-based system for translating queries into SQL and visualizing results using agentic workflows. [🔗](#)
- **Natural Language Inference Model Optimization:** Optimized RoBERTa and LLaMA models using LoRA and QLoRA for memory-efficient inference. [🔗](#)
- **Assessing Data Privacy in LLMs:** Evaluated Llama-3.2 against privacy threats including Membership Inference Attack and Jailbreaking (PAIR). [🔗](#)
- **Tumor Detection:** Detect and classify MRI images of the brain for the diagnosis of different brain tumors using a convolutional neural network. [🔗](#)
- **Machine Translation:** Compare LSTM and Transformer models for machine translation tasks. [🔗](#)
- **Customer Analytics:** Developed predictive models to identify customer trends while implementing privacy-preserving techniques to safeguard sensitive data. [🔗](#)
- **Selected Additional Projects:**
 - Monte-Carlo Localization & SLAM on Vector Robot [🔗](#)
 - Sentiment Analysis with N-grams [🔗](#)
 - Keep Fit: Human posture anomaly detection application. [🔗](#)
 - DAC: Deep Autoencoder-based Clustering. [🔗](#)